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Contraception

Contraception 63 (2001) 1–11
Original research article

Comparison of the lipoprotein, carbohydrate, and hemostatic effects of phasic oral contraceptives containing desogestrel or levonorgestrel

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Received 6 July 2000; revised 23 October 2000; accepted 21 November 2000

Abstract

Desogestrel (DSG) is a less-androgenic progestogen than levonorgestrel (LNG). This difference in androgenicity may be responsible for observed differences in metabolic effects between oral contraceptive (OC) formulations containing almost equivalent estrogen doses but with either DSG or LNG as a progestogen. To test the hypothesis, a prospective 9-month randomized comparison of plasma lipids, glucose, insulin, hemostasis, and sex hormone binding globulin (SHBG) was conducted in 66 healthy women using phasic formulations of OCs containing either DSG (DSG-OC) or LNG (LNG-OC). The study results showed that SHBG increased 3-fold with DSG-OC and 2-fold with LNG-OC. DSG-OC increased HDL-C, HDL_2-C and HDL_3-C; LDL-C decreased transiently. LNG-OC decreased HDL_2-C and increased HDL_3-C; HDL-C was unchanged and LDL-C decreased transiently. Both formulations increased VLDL-C and triglycerides, more with DSG-OC, but apolipoprotein B levels increased equally. Apo A-I and A-II increased more with DSG-OC than with LNG-OC. Neither formulation altered Lp(a) or fasting glucose and insulin levels. Postprandially, both formulations decreased glucose and increased insulin responses, but to an equivalent degree. Both OCs slightly enhanced procoagulant and profibrinolytic parameters to the same extent except for internally compensating decreases in Factor V and protein S with DSG-OC. In summary, at almost equivalent estrogen doses, a phasic OC containing DSG compared with LNG has a less androgenic effect on lipoproteins and SHBG, similar effects on hemostatic parameters with lower protein S and factor V activity and equivalent effects on carbohydrate metabolism. The lipoprotein, SHBG, and protein S and factor V differences are likely due to the lesser androgenicity of DSG allowing for a greater expression of the dose of estrogen. © 2001 Elsevier Science Inc. All rights reserved.

Keywords: Oral contraceptives; Desogestrel; Levornogestrel; Lipids; Hemostasis; SHBG; Carbohydrates

1. Introduction

Over the years, the use of combined oral contraceptives (COCs) has been associated with a few rare, but potentially serious side effects. In view of their rare occurrence, they cannot be properly investigated in clinical studies. Therefore, efforts have been made to define secondary parameters, which would provide some indication of the effect to be expected (surrogate endpoints). The choice of these surrogate endpoints was derived from associations found in men and post-menopausal women (lipids and arteriosclerosis) or from known associations between certain disease states and thrombosis (e.g., vascular complications in diabetes mellitus, deficiencies in certain clotting parameters associated with venous thrombosis). Although the predictive value of these parameters with respect to the occurrence of certain side effects in COC users has not been established, it has become good clinical practice to investigate the extent of metabolic disturbance caused by OCs. The early, high-dose combination OCs had marked effects on lipid and carbohydrate metabolism [1,2]. The hemostatic system was also affected [3]. Effects on plasma lipoproteins [4,5] were proportional to the relative biologic effects of the estrogen and progestogen constituents [6–9] as well as the inherent androgenicity of some of the OC progestogens [10–13]. Insulin resistance and glucose intolerance also

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PII: S0010-7824(00)00196-7

increased with progestogen dominance of the formulations [6,12]. With regard to hemostasis, estrogens have been associated with the risk of venous thrombosis [3,14–17] but there is no association between the type of progestogen and venous thrombosis [18].

The metabolic effects of high-dose COCs were initially reduced by reducing doses of estrogen and progestin and administering the progestin in a triphasic regimen, that is, in a step-wise, ascending manner from a lower initial dose [12]. With the introduction of more selective, new-generation progestogens with lower androgenicity, the metabolic effects of COCs have been further minimized [12,19] but the influence of different phasic regimens has still not been fully elucidated [12,20–25].

In the present report, we compare two phasic OC formulations containing almost equivalent doses of estrogen (665 μg ethinylestradiol (EE)/cycle for DSG-OC vs 680 μg EE/cycle for LNG-OC) and 2 progestogens in similar cumulative doses with different degrees of androgenicity; DSG, and the more androgenic LNG [10–13]. Effects on lipoprotein and carbohydrate metabolism and hemostasis were measured in a 9-month, randomized, parallel design study. This study addresses the practical question of how the effects of the DSG formulation on metabolic and hemostatic systems differ from those of the LNG formulation.

2. Materials and method

The recruitment of subjects to this protocol was reviewed and approved by the Human Subjects Review Committee of the University of Washington. Subjects eligible for enrollment were healthy women aged 21–35 years, with serum LDL cholesterol (LDL-C) and triglyceride levels lower than 160 and 250 mg/dL, respectively, and a regular menstrual cycle of 25–35 days. Subjects with illnesses affecting lipoprotein metabolism (i.e., hypothyroidism, liver disease, renal disease, or diabetes) were excluded. Additional exclusions included the use of OCs within 90 days or injectable hormonal preparations within 6 months or lipid-lowering agents or other drugs known to affect lipid or OC metabolism. Having hepatic, gastrointestinal, gynecologic, renal, hematologic, neurologic, respiratory, endocrine, or cardiovascular disease, a sitting diastolic blood pressure >90 mm Hg or systolic blood pressure >150 mm Hg were exclusionary, as were a history of smoking within 12 months, current use of tobacco or alcohol/drug abuse and pregnancy or breast-feeding within the past year. Subjects were to remain on a constant diet and make two visits to a dietician during the study to determine dietary stability.

At the first visit, written informed consent, medical histories and vital signs were obtained. The first study month comprised a normal baseline cycle, during which no OCs were taken. The baseline cycle visits were scheduled for days 9–11 and 22–24. During this month, physical and pelvic examinations were carried out and basal body temperatures were recorded, along with interim lifestyle, menses and drug histories. A 3-day food record was completed and analyzed for the amount and composition of dietary intake [26]. Blood samples were obtained on days 9–11 for derived determination of total plasma cholesterol (C), triglycerides, HDL-C, LDL-C [27], plasma sex hormone binding globulin (SHBG) [28] and human chorionic gonadotropin levels to exclude pregnancy. More extensive analyses were obtained on days 22–24 as described below.

Eligible subjects were randomly assigned to receive nine cycles of a phasic OC containing either DSG (50/100/150 μg) and EE (35/30/30 μg) in a 7/7/7-day regimen (Laurina®, NV Organon, The Netherlands {DSG-OC}) or LNG (50/75/125 μg) and EE (30/40/30 μg) in a 6/5/10-day regimen (Triphasil®-21, Wyeth-Ayerst Laboratories, Philadelphia, PA, USA {LNG-OC}). Subjects were instructed to take the first tablet on the first menstrual day following the normal baseline cycle. Tablets were taken at the same time each day, preferably at bedtime, for 21 consecutive days, followed by 7 pill-free days. Subjects initiated subsequent treatment courses on the same day of the week as the first.

SHBG levels were also determined during OC cycle 6, days 9–11 and 19–21. Vital signs were recorded, fasting lipoprotein measurements were obtained using the Friedewald estimation on days 19–21 of the first, third, and ninth OC cycles, and on days 9–11 and days 26–28 of OC cycle 6 [27]. More extensive estimations were made on days 22–24 of the baseline cycle and days 19–21 of OC cycle 6. These included direct determination of all lipid and apolipoprotein parameters (fasting total cholesterol, triglycerides, LDL-C, VLDL-C [27], HDL-C [29] and subfractions HDL_2-C and HDL_3-C [30], apolipoproteins Apo A-I [31], A-II [32], B [33], Lp(a) [34], and SHBG [28]). Also, glucose [35] and insulin levels [36] were measured fasting and one hour after ingestion of a 400-calorie glucose/protein/fat drink. Plasma triglyceride and VLDL-C were also measured one hour after the drink. Analyses on hemostatic parameters included assays for prothrombin time (PT) [37], activated partial thromboplastin time (APTT) [38], fibrinogen [37], [38], factors V [39], and VIII [40], antithrombin III [41], protein C activity [42], protein S antigen [43], tissue plasminogen activator (t-PA) antigen [44], and plasminogen activator inhibitor type 1 (PAI-1) activity [45,46]. Clotting assays were performed in the Department of Laboratory Medicine, University of Washington, School of Medicine.

2.1. Statistical analysis

Statistical comparison of data obtained at baseline and at cycles 1, 3, 6, and 9 for the two study drugs employed the 2-sample *t* test or the non-parametric analog Wilcoxon rank-sum test. Fisher's exact test or chi-square test was used to assess the categorical variables. For statistical analysis of the HDL-C/total cholesterol ratio and HDL-C/LDL-C ratios in the first cycle, log-transformed data were used.

Table 1
Characteristics of the study population

	DSG-OC (n = 32)	LNG-OC (n = 34)
Age (years)[a]	28.1 ± 3.8	27.6 ± 3.7
Race[b]		
White	31 (96.9)	32 (94.1)
Black	0 (0.0)	1 (2.9)
Asian/Oriental	1 (3.1)	1 (2.9)
Parity[b]		
0	20 (62.5)	29 (85.3)
1	5 (15.6)	2 (5.9)
2	6 (18.8)	3 (8.8)
3	1 (3.1)	0 (0.0)
Duration of priori OC use (years)[b]		
0	3 (9.4)	1 (2.9)
<1	1 (3.1)	0 (0.0)
1–5	11 (34.4)	20 (58.8)
>5	17 (53.1)	13 (38.2)
Systolic blood pressure (mm Hg)[a]	100.3 ± 12	103.4 ± 11.5
Diastolic blood pressure (mm Hg)[a]	63.0 ± 7.6	63.3 ± 7.4
Body mass index (kg/m^2)[a]	22.4 ± 2.4	23.1 ± 4.0
Dietary intake[a]		
Kilocalories/day	1877.0 ± 454.0	1803.0 ± 470.0
Cholesterol (g/day)	243.0 ± 108	213.0 ± 92
Fiber (g/day)	15.4 ± 6.6	15.1 ± 6.2
Fat (% of total calories)	34.3 ± 5.1	35.1 ± 6.3
SFA (% of total calories)	12.6 ± 2.4	12.3 ± 3.0

[a] Mean ± SD.
[b] Number (%) of subjects.

3. Results

3.1. Subject characteristics

Thirty-three healthy female subjects were randomized to receive DSG-OC and 34 to LNG-OC (Table 1). One subject in the DSG-OC group became pregnant during the baseline cycle and did not take the study medication; she was excluded from the analyses. The study population was predominantly white and nulliparous, and 80% of the women were over 25 years of age. The women were lean on average and body mass index was comparable between the two groups. Most of the women had a history of previous OC use of at least 1 year. No significant differences were detected in race, parity, previous OC use, blood pressure, body weight or dietary fat and energy intake.

3.2. Clinical observations

No pregnancies occurred in either group during the study. Daily diaries indicated that withdrawal bleeding was comparable between the two groups. No subjects discontinued treatment due to menstrual problems or dysmenorrhea. One subject in the DSG-OC group and three in the LNG-OC group discontinued treatment prematurely for noncompliance or other protocol violations. Body weight, blood pressure and fat intake scale values are shown in Table 1. Body mass index did not differ significantly between the two groups; there was a slight increase in both groups that was more pronounced with LNG-OC (Table 2). A slight but significant rise in systolic blood pressure was seen only in cycle 9 in DSG-OC compared with LNG-OC ($p = 0.022$). No difference in diastolic blood pressure was seen. No statistically significant differences from baseline or between the DSG-OC and LNG-OC groups were seen in the fat intake scale.

3.3. Effects on SHBG levels

Mean SHBG levels increased in both treatment groups during the baseline cycle and over the first 6 months of the treatment period (Table 3). During the baseline cycle, mean SHBG levels on days 22–24 were 8% and 7% higher than those on days 9–11 in the DSG-OC group and LNG-OC groups, respectively (significant in the DSG-OC group). At cycle 6, SHBG levels in the DSG-OC group were about 3- and 3.5-fold higher than the corresponding early- and late-phase baseline values (Table 3). In contrast, the LNG-OC

Table 2
Body mass index and blood pressure values (mean ± SD) at baseline and changes with OC use

	Baseline cycle (day 22–24)	Change from baseline (day 19–21)			
		Cycle 1	Cycle 3	Cycle 6	Cycle 9
Number of subjects					
DSG-OC	32	32	30	30	26
LNG-OC	34	27	27	27	26
BMI (kg/m^2)					
DSG-OC	22.4 ± 2.4	0.06 ± 0.50	0.23 ± 0.61	0.42 ± 0.80	0.49 ± 1.07
LNG-OC	23.1 ± 4.0	0.16 ± 0.42	0.40 ± 0.69	0.77 ± 1.01	0.95 ± 1.71
Systolic blood pressure (mm Hg)					
DSG-OC	100.3 ± 12.5	3.5 ± 12.5	1.9 ± 10.7	3.2 ± 12.6	8.0 ± 11.6
LNG-OC	103.4 ± 11.5	−2.8 ± 11.5	−0.1 ± 11.2	0.8 ± 13.1	0.3 ± 13.6
Diastolic blood pressure (mm Hg)					
DSG-OC	62.0 ± 7.6	−0.2 ± 7.6	−0.3 ± 7.8	−0.1 ± 1.1	3.3 ± 8.9
LNG-OC	63.3 ± 7.4	0.3 ± 7.4	−2.5 ± 8.5	1.8 ± 7.9	1.7 ± 7.5

Table 3
Effects of OC formulations on plasma SHBG levels (nmol/L) in the baseline cycle and cycle 6

	Baseline cycle		Cycle 6	
	Day 9–11	Day 22–24	Day 9–11	Day 19–21
DSG-OC				
Mean	52.5	56.9	167.2	186.7
Change (mean ± SD)	—	4.5 ± 7.5[a]	114.7 ± 57.4[a]	129.8 ± 72.1[a,b]
LNG-OC				
Mean	49.6	53.3	109.2	103.2
Change (mean – SD)	—	3.7 ± 11.8	59.1 ± 23.2[a]	49.9 ± 30.1[a,b]
p-value[c]	NS	NS	<0.001	<0.001

[a] Significant ($p < 0.05$) within-treatment group difference from baseline.
[b] Significant ($p < 0.05$) increase from comparable time in menstrual cycle.
[c] Comparison of SHBG increases with DSG-OC and LNG-OC use from the corresponding time in the normal menstrual cycle.

formulation was associated with approximately 2-fold elevations early and late in the 6th cycle over baseline. All increases above baseline were significant, as were the greater increments of the DSG-OC group over the LNG-OC group. Between days 9–11 and days 19–21 of cycle 6, there was a further SHBG increase in the DSG-OC group ($p < 0.05$) but not the LNG-OC group.

3.4. Evaluation of lipid and lipoprotein parameters

Fasting lipoprotein lipid and apoprotein levels changed in both treatment groups between baseline to cycle 9. As seen in Table 4, the total plasma cholesterol level increased significantly from baseline at cycles 3, 6, and 9 in the DSG-OC group and at cycle 6 in the LNG-OC group. No significant differences were seen between the 2 OC groups.

Plasma triglyceride levels increased significantly from baseline in both treatment groups at all cycles. At cycles 1, 6, and 9, the increase in the DSG-OC group was statistically significantly higher compared with the LNG-OC group (Fig. 1). Similarly, VLDL levels increased significantly at all evaluated cycles in the DSG-OC group and in 3 of the cycles in the LNG-OC group (Table 4). However, the VLDL levels in DSG-OC users significantly exceeded those in LNG-OC users at all visits. Significant decreases in LDL-C levels were detected at cycles 1 and 3 in the DSG-OC group and at cycle 1 in the LNG-OC group. Thereafter, LDL-C trended upward in both groups at cycle 6, significantly so in the LNG-OC group. LDL-C levels were lower in the DSG-OC vs LNG-OC group at all OC cycle visits, reaching statistical significance at cycles 1 and 3.

HDL-C levels were significantly increased at all DSG-OC cycle visits (Table 4 and Fig. 1). Levels of both HDL_2-C and HDL_3-C, measured at cycle 6, were also increased (Table 4). As a result, the HDL-C/total cholesterol ratio for the DSG-OC group was significantly increased at all cycle visits (Table 4 and Fig. 1). In contrast, HDL-C did not change in the LNG-OC group, reflecting a statistically significant decrease in HDL_2-C and a smaller increase in HDL_3-C (compared with the DSG-OC group). These changes resulted in a statistically significantly lower HDL-C/total cholesterol ratio in the LNG-OC group at all visits compared with the DSG-OC group.

Values for both Apo A-I and A-II (Table 4) were significantly increased in both DSG-OC and LNG-OC groups at cycle 6, but significantly more so in the former when compared with baseline levels. Apo B increased in both groups to a similar extent. Plasma Lp(a) levels did not change during use of DSG-OC or LNG-OC. The major differences in lipoprotein, lipid and apoprotein changes seen between the 2 formulations are summarized in Fig. 2.

3.5. Effects of OC use on carbohydrate and lipid metabolism after a 400-kcal mixed liquid meal

No significant changes from baseline to visit 6 were observed in fasting glucose and insulin levels in either treatment group (Table 5). One hour after ingestion of a 400-calorie protein/carbohydrate/fat drink, plasma glucose levels were significantly below fasting levels in both groups at both baseline and cycle 6, whereas insulin levels were correspondingly higher. Both OC formulations tended to blunt the one-hour decrease in glucose. No significant differences between the two groups were seen in baseline fasting glucose and insulin levels nor in the changes from baseline. Likewise, no significant differences were observed between the formulations in the increments in triglyceride and VLDL-C in the baseline and OC cycles.

3.6. Effect on coagulation and fibrinolytic parameters

Changes in some coagulation parameters between baseline and cycle 6 were observed in both treatment groups (Table 6), some favoring and some opposing coagulation. Favoring coagulation were slight decreases in PT and APTT in both treatment groups and increases in fibrinogen and factor VIII, again in both groups (significantly for factor VIII only in the DSG-OC group), and a decrease in free and total protein S, only in the DSG-OC group. A change

Table 4
Fasting plasma lipoprotein changes (mean ± SD)

	Baseline cycle (days 22–24)	Change from baseline (days 19–21)			
		Cycle 1	Cycle 3	Cycle 6	Cycle 9
Total cholesterol (mg/dL)					
DSG-OC	171.6	−2.0 ± 23.1	−12.13 ± 19.9[a]	24.4 ± 23.8[a]	18.4 ± 30.7[a]
LNG-OC	175.4	−3.2 ± 19.8	0.2 ± 24.9	11.4 ± 28.9[a]	10.9 ± 28.1
p-value		NS	NS	NS	NS
Triglyceride (mg/dL)					
DSG-OC	57.9	34.1 ± 24.1[a]	47.5 ± 36.9[a]	47.2 ± 22.4[a]	44.5 ± 32.2[a]
LNG-OC	58.8	11.5 ± 27.8[a]	18.0 ± 26.7[a]	16.3 ± 24.4[a]	21.2 ± 24.9[a]
p-value		<0.002	NS	<0.001	<0.007
VLDL-C (mg/dL)					
DSG-OC	11.6	9.5 ± 6.3[a]	12.3 ± 8.4[a]	6.8 ± 7.7[a]	11.7 ± 6.8[a]
LNG-OC	11.8	5.0 ± 7.1[a]	6.2 ± 7.4[a]	1.6 ± 6.9	7.1 ± 6.4[a]
p-value		0.01	0.004	0.007	0.01
LDL-C (estimated) (mg/dL)					
DSG-OC	101.5	−18.3 ± 18.4[a]	−9.5 ± 16.8[a]	6.4 ± 20.4	−3.6 ± 20.4
LNG-OC	100.8	−8.1 ± 19.1[a]	−4.5 ± 20.3	9.3 ± 23.1[a]	4.5 ± 22.7
p-value		0.06	0.06	NS	NS
HDL-C (mg/dL)					
DSG-OC	58.5	6.6 ± 7.4[a]	9.3 ± 7.8[a]	11.0 ± 7.4[a]	10.0 ± 8.3[a]
LNG-OC	62.8	−0.2 ± 6.4	−1.7 ± 7.5	0.3 ± 9.9	−0.7 ± 11.8
p-value		NS	<0.001	NS	NS
HDL_2-C (mg/dL)					
DSG-OC	20.2	—	—	2.2 ± 4.1[a]	—
LNG-OC	23.8	—	—	−4.4 ± 5.6[a]	—
p-value				<0.001	
HDL_3-C (mg/dL)					
DSG-OC	38.3	—	—	8.7 ± 5.2[a]	—
LNG-OC	39.0	—	—	4.5 ± 5.3[a]	—
p-value				0.003	
HDL-C/Total cholesterol ratio					
DSG-OC	0.34	0.043 ± 0.036	0.030 ± 0.035	0.015 ± 0.040	0.025 ± 0.058
LNG-OC	0.36	0.006 ± 0.045	−0.003 ± 0.043	−0.016 ± 0.040	−0.019 ± 0.055
p-value		0.02	0.02	0.004	0.008
Apo A-I (mg/dL)					
DSG-OC	152.7	—	—	39.0 ± 18.7[a,b]	—
LNG-OC	156.7	—	—	14.9 ± 21.1[a]	—
p-value				<0.001	
Apo A-II (mg/dL)					
DSG-OC	23.4	—	—	5.6 ± 3.1[a]	—
LNG-OC	25.3	—	—	3.0 ± 3.8[a]	—
p-value				<0.05	
Apo B (mg/dL)					
DSG-OC	77.5	—	—	15.6 ± 14.5[a]	—
LNG-OC	77.5	—	—	15.5 ± 18.1[a]	—
p-value				NS	

[a] Significantly different from baseline ($p < 0.05$).
[b] Significantly different from mean change in LNG-OC group ($p \leq 0.03$).

favoring diminished coagulation was a reduction in factor V in the DSG-OC group only and increases in protein C in both groups. Both groups showed reductions in t-PA antigen levels (favoring reduced fibrinolysis) and PAI-1 activity (favoring increased fibrinolysis), with a slightly larger reduction observed in the LNG-OC group. No significant change was seen in either OC formulation for antithrombin III. Responses were significantly divergent between the two formulations only for factor V and free and total protein S as also shown in Fig. 2.

4. Discussion

The purpose of the present study was to compare the effects on lipid and carbohydrate metabolism and on hemostasis of two phasic OC formulations containing EE at nearly equivalent doses and DSG or LNG. Less androgenic effects in the DSG-OC group were most clearly seen in the higher SHBG, total triglyceride, and VLDL cholesterol levels, a trend toward lower LDL-C levels and higher HDL-C levels at all time points over 9 months compared with

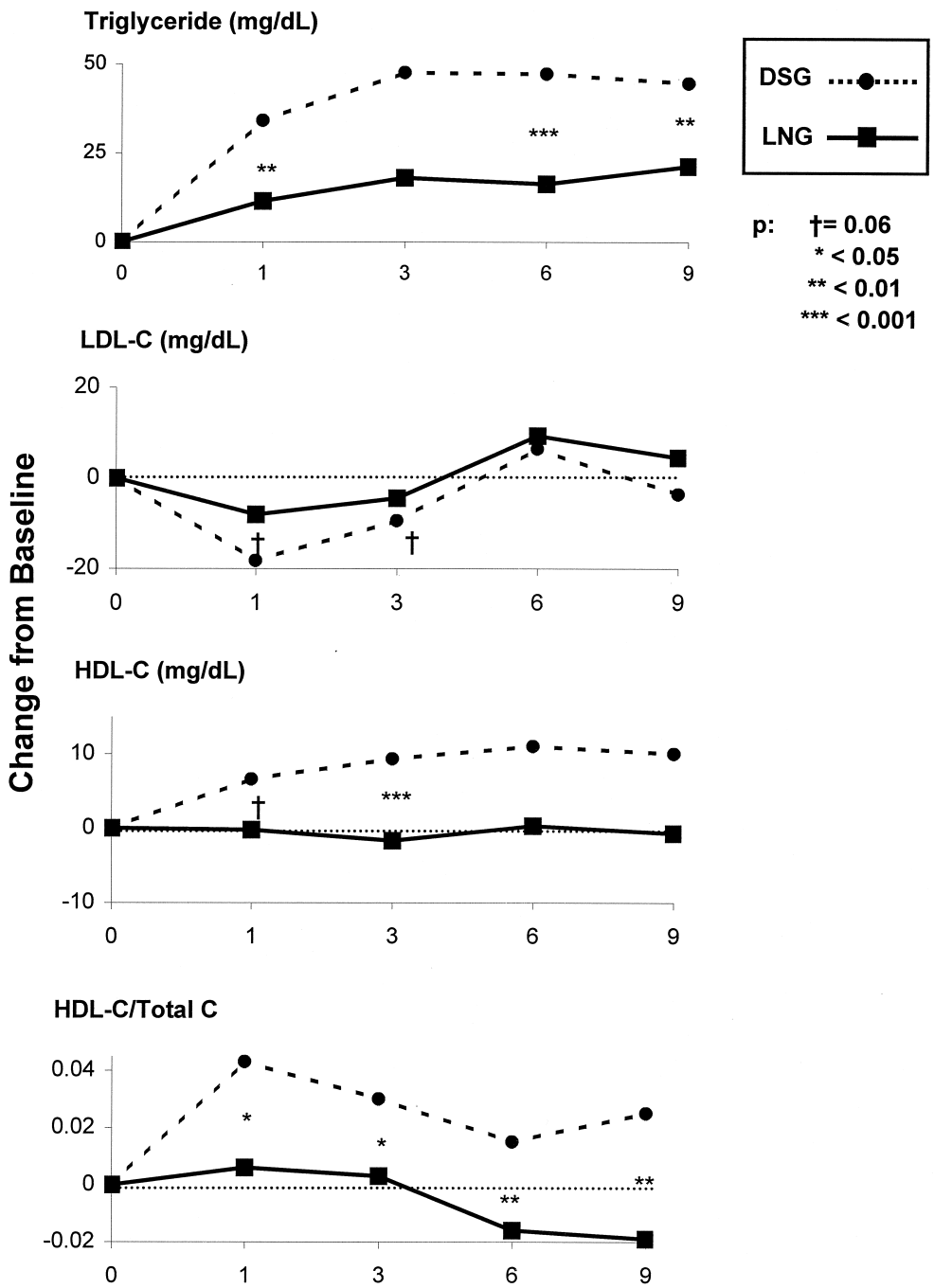

Fig. 1. Effects of DSG-OC and LNG-OC on the major lipoprotein lipids over the 9 months of the study.

LNG-OC. In addition, HDL_2-C and HDL_3-C and apoproteins A-I and A-II were all significantly higher with DSG-OC compared with LNG-OC after 6 months. Similar results have been reported by Cheung et al., who found a less androgenic effect of DSG-OC in immunochemically separated subclasses of HDL containing apo A-I only or apo A-I and A-II together in a subset of subjects studied from this cohort [47].

The present results show a slightly less androgenic effect compared with the published experience with monophasic DSG/EE formulations [12]. Twenty-three studies showed a 29% increase in triglycerides, a 3% increase in cholesterol, a 2% decrease in LDL-C, and a 13% increase in HDL-C [12]. Apoproteins A-I and B both increased by 11%. Compared with that of the monophasic DSG/EE profile, the phasic DSG-containing formulation would be expected to be less androgenic due to the one-third reduction in the dose of the progestogen. This prediction is borne out by an 81%

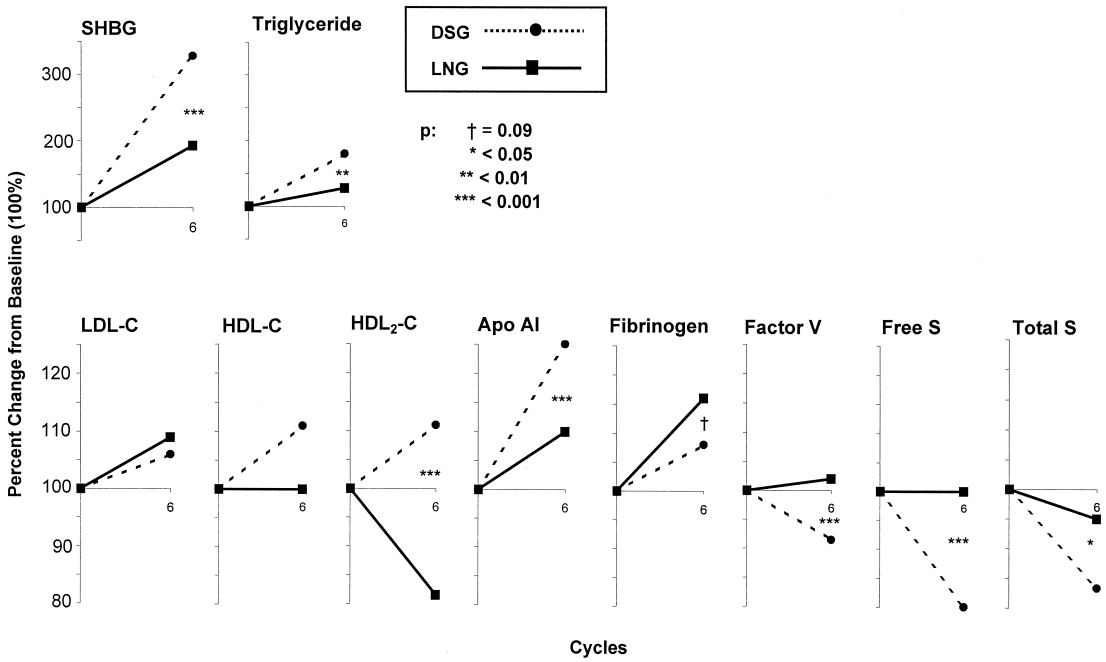

Fig. 2. Effects of DSG-OC and LNG-OC on SHBG, lipoprotein and clotting parameters showing sensitivity to differences in estrogen/progestogen balance in the 2 formulations. Data shown are percentage increases or decreases from late luteal phase normal menstrual cycle (arbitrarily set to 100%). SHBG and certain lipoprotein and clotting factors show differences in estrogen-progestogen balance between women taking DSG-OC or LNG-OC (6-month data). Statistics are based on the comparison of the increments in absolute values as shown in Tables 2, 4, and 6. Though HDL-C changes were not significantly different at 6 months as shown, differences were significant at 0 and 3 months, a trend that was continued at 6 and 9 months.

increase in triglycerides, a 19% increase in HDL-C and a 26% increase in apo A-I over the late baseline cycle values (Fig. 2).

With respect to the lipoprotein changes seen with the LNG-OC formulation, our present results confirm three previous observations [48–50]: we found a 27.5% increase in triglyceride, a 5% increase in cholesterol, a 7% increase in LDL-C, and no change in total HDL-C with a 20%

Table 5
Comparison of mean fasting values and one-hour changes from baseline in carbohydrate and lipid parameters after a 400 Cal breakfast drink (mean ± SD)

	Baseline cycle (days 22–24)		Cycle 6 (days 19–21)	
	Fasting	1-h difference from baseline	Fasting	1-h difference from baseline
Glucose (mg/dL)				
DSG-OC	86.1	−13.4 ± 14.0[a]	86.0	−6.2 ± 15.1[a]
LNG-OC	84.2	−10.8 ± 19.7[a]	87.3	−8.1 ± 22.2
p-value		NS		NS
Insulin (μU/ml)				
DSG-OC	14.6	45.1 ± 25.8[a]	15.8	60.5 ± 25.6[a]
LNG-OC	14.7	47.4 ± 24.6[a]	17.0	47.9 ± 27.6[a]
p-value		NS		NS
Triglyceride (mg/dL)				
DSG-OC	57.9	25.2 ± 19.8[a]	104.0	23.9 ± 20.4[a]
LNG-OC	58.8	17.9 ± 11.1[a]	75.0	23.4 ± 13.5[a]
p-value		NS		NS
VLDL-C (mg/dL)				
DSG-OC	9.0	7.7 ± 5.0[a]	15.5	10.2 ± 5.9[a]
LNG-OC	9.1	6.2 ± 4.3[a]	10.8	9.0 ± 4.9[a]
p-value		NS		NS

p-value compares the 1-hour difference from baseline for the two formulations.
[a]p <0.05 for the significance of 1-h difference from baseline.

Table 6
Changes in hemostatic parameters

	Baseline cycle (days 22–24)	Cycle 6 (days 19–21)	Change from baseline	p-value[b]
Overall clotting time(1+2)				
[1]Prothrombin time (sec)				
DSG-OC	12.1	11.9[a]	−0.2 ± 0.3	NS
LNG-OC	12.2	11.9[a]	−0.3 ± 0.4	
[2]Activated prothrombin time (sec)				
DSG-OC	28.5	27.7[a]	−0.8 ± 2.0	NS
LNG-OC	28.4	27.4[a]	−1.0 ± 1.8	
(Pro)coagulatory factors				
Fibrinogen (mg/dL)				
DSG-OC	262.6	282.5[a]	19.0 ± 50.9	0.09
LNG-OC	268.5	311.6[a]	43.1 ± 55.2	
Factor V (%)				
DSG-OC	97.2	88.8[a]	−8.4 ± 11.4	<0.001
LNG-OC	95.2	97.3	2.2 ± 13.1	
Factor VIII (%)				
DSG-OC	95.2	106.7[a]	11.5 ± 21.2	NS
LNG-OC	97.1	104.7	7.6 ± 26.1	
Anticoagulatory parameters				
Antithrombin III (%)				
DSG-OC	94.5	94.5	0.1 ± 6.9	NS
LNG-OC	97.6	99.5	1.9 ± 10.0	
Protein C (%)				
DSG-OC	96.1	116.0[a]	19.9 ± 8.3	NS
LNG-OC	98.5	114.5[a]	16.0 ± 14.3	
Free protein S (U/mL)				
Free protein S (U/mL)				
DSG-OC	1.0	0.8[a]	−0.2 ± 0.2	<0.001
LNG-OC	1.1	1.1	0.0 ± 0.2	
Total protein S (U/mL)				
DSG-OC	2.4	2.0[a]	−0.4 ± 0.7	0.047
LNG-OC	2.5	2.4	−0.1 ± 0.4	
(Pro)fibrinolytic parameter				
t-PA antigen (ng/mL)				
DSG-OC	4.1	2.6[a]	−1.5 ± 1.6	NS
LNG-OC	5.0	2.8[b]	−2.1 ± 1.6	
Antifibrinolytic parameter				
PAI-1 activity (AU/mL)				
DSG-OC	4.6	3.3[a]	−1.4 ± 3.3	NS
LNG-OC	6.8	3.2[a]	−3.6 ± 7.2	

[a]Significantly different from baseline ($p < 0.05$).
[b]Significantly different from mean change in LNG-OC group.

decrease in HDL_2-C and a 12% increase in HDL_3-C. The 3 previous reports [48–50]. showed a triglycerides increase in the range of 28–31%, an LDL-C increase of 6–11%, and an HDL-C decrease of 1–2% (see references 7–9 for reviews).

The impact of an androgenic progestogen in an OC formulation is most apparent from the changes in SHBG levels, an accepted index of estrogen/androgen effect. SHBG increased over 3-fold above the baseline cycle with the DSG-OC and only 2-fold with the LNG-OC formulation. Similar differences were observed when monophasic DSG and LNG formulations were compared [51].

SHBG changes during the 6th cycle similarly reflect differences in hormone exposure. In the DSG-OC group, plasma SHBG concentration which was already high at the beginning of the cycle, increased even further later in the OC cycle despite a rising dose of DSG, possibly reflecting the longer exposure to EE near the end of the cycle. In contrast, LNG-OC was associated with a slight reduction in SHBG late in the OC cycle despite a similar or greater exposure to EE, consistent with a more pronounced androgenic effect as a result of increasing doses of LNG.

With respect to the effects on carbohydrate metabolism, previous experience indicates that insulin resistance and glucose intolerance are associated with the use of androgens or androgenic progestogens [7,52], while estrogen use is associated with a decrease in plasma glucose and insulin concentrations in post-menopausal women [53,54]. In contrast to the differential effects on SHBG and lipoprotein levels, no significant differences between the formulations were noted in either fasting or postprandial measurements,

including a slight one-hour postprandial reduction in plasma glucose concentration in both groups. This lower glucose value may be related to the rapid secretion of insulin in response to the protein-carbohydrate-fat-containing drink. Estrogen/progestogen balance appears to have a larger impact on hepatic lipoprotein metabolic pathways than pancreatic islet insulin secretion or insulin receptor activity.

Lp(a) is also insensitive to the differences in estrogen/progestogen-androgen balance inherent in DSG-OC and LNG-OC. Apo(a), a component of Lp(a), is a plasminogen analog with anti-plasminogen activity associated with arteriosclerotic risk [7]. At least one previous study of DSG- and LNG-containing formulations has been published showing no effect on Lp(a) [55]. Another report found a transient decrease in Lp(a) at one and 3 months of use of a biphasic DSG-containing OC [56]. These observations are especially surprising in light of reports that an androgen, estrogen, and progestogen all taken alone are associated with reductions in Lp(a) [53,57,58]. More work is needed to understand the regulation of Lp(a) and its biologic effect in complex hormonal systems.

Regarding OC effects on hemostasis, two general conclusions can be drawn. First, coagulatory effects induced by both formulations (decreased PT and APTT, increased fibrinogen and a decrease in the fibrinolytic parameter, t-PA antigen) appear to be offset by an anticoagulatory effect (increased protein C) and decrease of the antifibrinolytic parameter, PAI-1 activity. These are presumably estrogen-induced effects inherent in both OC formulations. On the other hand, differential sensitivity was seen in three of the 11 measured hemostasis parameters. Decreased factor V and decreased free and total protein S were observed with DSG-OC while LNG-OC showed no effect. The two effects suggest an opposing influence, i.e., reductions in both procoagulation and anticoagulation factors. These changes support the idea that some estrogen-related changes in clotting factors can be modulated by the androgenic effects of the progestogen. These changes with DSG-OC are also consistent with a recent review of the new OC progestogens [12] describing an increase in fibrinogen, no change in antithrombin III and protein C, and decreases in protein S. The present study confirms and extends these results and observes that some clotting factors are sensitive to progestogen as well as estrogen effects, confirming recent predictions of such an effect [59–62].

To conclude, phasic OC formulations containing DSG or LNG at almost equivalent EE doses have been studied for effects on lipid, (lipo)protein, carbohydrate, and hemostatic parameters. The DSG-OC formulation is less androgenic than the LNG-OC formulation on a ratio of 3:2 based on differential SHBG increases in the two formulations. This lower androgenicity is confirmed with the higher triglycerides, HDL, HDL_2-C, HDL_3-C and Apo A-I and A-II levels and slightly lower LDL-C levels seen with the DSG-OC compared with the LNG-OC formulation. Differences in clotting factor alterations were fewer, seen in only three of the 11 parameters studied. No differences were seen in fasting and postprandial glucose and insulin levels, nor in postprandial triglycerides and VLDL-C levels. These data underscore the sensitivity of SHBG and lipoprotein levels to fluctuations in estrogen and progestogen balance but also indicate that in selected parameters (protein S and factor V), distinctions in clotting factor levels may exist as well, suggesting that along with estrogens, progestogens also affect some clotting factors.

Acknowledgments

We thank Mr. George Bargoud and Dr. Tom Brakel for assistance. This work was supported by a grant from Organon Inc. and by the NIH Clinical Nutrition Research Unit (CNRU) at the University of Washington, DK-35816, NIH Program Project Grant HL-30086, and a generous gift from the Robert B. McMillen Family Trust.

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